Discretization as Synchronization: Quantum Spectra Without Quantizing Spacetime

A Mechanistic Completion of Penrose's Modify-QM-Not-GR Program

John Bramble, MD

May 2026

Contents

John Bramble, MD[1]

[1] Independent Research

Correspondence: rayolddog (GitHub)

Preprint — not yet peer reviewed

AI Disclosure: This work was developed in collaboration with Claude Opus 4.7 (Anthropic), as described in the Author Contributions section. Per current journal guidelines, LLMs do not satisfy authorship criteria; the human author bears full responsibility for all content.

Companion paper to: Bramble, J. (2026), "Many Clocks Interpretation of Quantum Mechanics: Mass as Chiral Coupling, Re-synchronization in the Bulk as Measurement" [PAPER_UNIFIED, under review].

Abstract

The standard formulation of quantum mechanics treats discreteness — discrete spectra, discrete spin values, discrete detection events — as a foundational feature requiring postulation. General relativity, in contrast, is formulated on a continuous, differentiable manifold. The tension between these two ontologies has driven nearly a century of attempts to either quantize spacetime (loop quantum gravity, causal sets, string theory, causal dynamical triangulations) or to derive a discrete substructure from a more fundamental theory. None has produced a confirmed prediction.

This paper proposes that the tension is misdirected. The discreteness of quantum mechanics is not a fundamental feature of nature but an *emergent property of synchronization on a continuous substrate*. Discrete spectra are the locked-mode spectra of nonlinear oscillator dynamics; discrete detection events

are the threshold-crossing kinematics of bulk-coupled detectors; discrete spin values are the irreducible-representation dimensions of an underlying continuous symmetry group. In each case the substrate remains continuous and the discreteness is derived. If this reading is correct, the manifold of general relativity need not be quantized — and the GR/QM conflict, as traditionally posed, dissolves.

The proposal builds on the Dirac–Kuramoto (DK) framework developed in the companion paper [1], which identifies the chiral structure of the Dirac equation as a phase-synchronization system and re-interprets measurement as a two-stage re-synchronization process. Here we extract the broader consequence: that the mechanism developed there for measurement generalizes to a *uniform mechanism for QM discreteness*, of which the standard examples (Bohr–Sommerfeld quantization, spin multiplets, atomic line spectra, photodetection, energy bands, cavity quantization, black-hole ringdown spectra) are all special cases. The position aligns with Penrose's long-standing argument [2, 3] that general relativity is correct and quantum mechanics requires modification; the DK mechanism is the dynamical content that Penrose's gravitational-collapse threshold left unspecified. The paper is an *interpretive proposal*, not a complete theory; the load-bearing derivations — particularly the basin-of-attraction measure that should recover the Born rule, and the high-precision spectrum of hydrogen-like atoms from a strict sync derivation — are identified as next steps.

---

1. Introduction

1.1 The traditional framing of the GR/QM conflict

General relativity describes spacetime as a smooth pseudo-Riemannian manifold whose curvature is sourced by stress-energy. The dynamical variables are continuous fields: the metric $g_{\mu\nu}(x)$, the connection, the Riemann tensor. Continuity is built in at the level of the formalism; discontinuities (shocks, singularities) are pathologies, not features.

Quantum mechanics, by contrast, exhibits discreteness everywhere one looks. Bound-state spectra of the hydrogen atom take discrete energy values $E_n = -13.6\ \mathrm{eV}/n^2$. Angular momentum eigenvalues are integer or half-integer multiples of $\hbar$. Spin-$\frac{1}{2}$ measurements yield two outcomes; spin-1 yields three. Photons arrive at detectors as discrete clicks regardless of how slowly the source is operated. Bose–Einstein and Fermi–Dirac statistics, the Pauli exclusion principle, the Aharonov–Bohm phase, the quantum Hall plateaus — all are statements about discreteness in a system whose underlying Schrödinger or Dirac equation is itself a smooth partial differential equation.

The traditional reading of this asymmetry is that the discreteness is fundamental — that nature is "actually" discrete at small scales and the continuous formalism of GR is the macroscopic approximation. The natural program is then to quantize spacetime: introduce a fundamental length scale (the Planck length $\ell_P \approx 1.6 \times 10^{-35}$ m) and build an underlying discrete structure from which the smooth GR manifold emerges in the long-wavelength limit. Loop quantum gravity [4] derives discrete area and volume operators with non-zero minimum eigenvalues; causal-set theory [5] takes a discrete causal partial order as foundational; causal dynamical triangulations [6] discretize spacetime explicitly; string theory [7] introduces a fundamental string length below which spacetime ceases to be locally Minkowski.

After roughly a half-century of intensive work, none of these programs has produced a confirmed experimental prediction distinct from standard GR or standard QM. The cosmological-constant problem remains unsolved within each. The fundamental obstacles are remarkably consistent: each program must recover *smooth* GR in the classical limit (which requires that the discreteness almost-but-not-quite cancel itself out at macroscopic scales), and each must recover *standard* QM at experimentally accessible energies (which requires that the discrete substructure not contaminate well-tested quantum predictions). The combined constraints are severe.

1.2 Penrose's alternative position

A minority position, defended most prominently by Roger Penrose [2, 3], is that the traditional framing has the asymmetry backwards: it is not GR that needs to be modified to accommodate QM, but QM that needs to be modified to accommodate GR. The linear unitary evolution of quantum mechanics — the Schrödinger equation taken as exact — is what should give way, while the smooth differentiable manifold of GR is preserved. Penrose's objective reduction (OR) proposal supplies a specific mechanism: a mass-energy superposition collapses on a timescale $\tau \sim \hbar/E_G$, where E_G is the gravitational self-energy of the difference between the superposed mass configurations. The collapse is real, gravitational in origin, and breaks linearity at a calculable threshold.

OR has not been confirmed (current experimental bounds [8, 9] are consistent with but do not require the prediction), and the proposal has two acknowledged structural gaps. First, the threshold is stipulated rather than derived: the formula $\tau \sim \hbar/E_G$ comes from a plausibility argument about the energy cost of maintaining gravitationally-distinct superpositions, not from a dynamical equation. Second, the proposal explains *when* collapse happens but not *why* it produces specific discrete outcomes — the eigenstate selection mechanism is left to standard QM, which is the very thing being modified.

This paper proposes that both gaps close together if the underlying picture is correct.

1.3 The proposal

The proposal is this: the discreteness of quantum mechanics is not a fundamental feature of nature but an emergent consequence of *phase synchronization on a continuous substrate*. Specifically:

1. The underlying field dynamics — the Dirac equation for fermions, the Maxwell equations for photons, the linearized Einstein equations for weak gravity — are continuous partial differential equations on a smooth manifold. None of them are intrinsically discrete.
2. These continuous dynamics admit nonlinear *synchronization* dynamics when the field modes interact, of the type studied extensively by Kuramoto and his successors [10]. The synchronization equations are themselves continuous, but they possess a *discrete spectrum of locked modes* — phase-coherent attractors selected by basin geometry.
3. The discreteness observed in quantum mechanics is the discreteness of these locked modes, not of the substrate. Discrete spectra are eigenvalues of synchronization operators on the locked manifold; discrete detection events are the threshold-crossings of bulk-coupled detectors; discrete spin values are the dimensions of irreducible representations of the continuous symmetry group acting on the substrate.
4. Consequently, general relativity's smooth manifold need not be quantized. Every instance of QM discreteness has a sync-based origin *within the matter sector*; none requires a discrete substructure in spacetime itself.

The proposal is in agreement with Penrose's position — that GR is correct and QM is what gives way — but it supplies the dynamical mechanism that Penrose's threshold formula left unspecified. The mechanism is the two-stage Kuramoto re-synchronization developed in the companion paper [1]. Penrose's $\tau \sim \hbar/E_G$ emerges as a particular limit of that mechanism in the gravitational regime; the broader claim — that *all* of quantum discreteness is sync-emergent — is a generalization that Penrose did not pursue.

1.4 What this paper does and does not claim

This paper makes one claim: that the discreteness of quantum mechanics is synchronization-emergent on a continuous substrate, and that consequently the GR/QM conflict (in the traditional "either spacetime must be quantized or QM must be modified" framing) is resolved by modifying QM in the specific direction the DK framework prescribes.

The paper does not claim to derive the Standard Model coupling constants, to predict new particles, or to compute corrections to known QED results. It does not claim to derive the hydrogen spectrum to spectroscopic precision from first principles in the sync language (this is an acknowledged open task — §6). It does not propose new dynamics; the mechanism is what is developed in [1]. Its contribution is *interpretive relocation*: showing that a single mechanism, already required by the companion framework for measurement, accounts for the broader phenomenon of QM discreteness, and that this relocation dissolves the standard GR/QM tension.

It is also not a competitor to loop quantum gravity, string theory, or causal-set programs in their main aims. Those programs ask: what is the quantum theory of the gravitational field? The DK proposal answers a different question: *why does QM look discrete to begin with?* If the answer is "sync emergence on a continuous substrate," then the question those programs address — how to quantize the gravitational field — may not need an affirmative answer, because the discreteness it presupposes on the matter side is itself derived.

---

2. The Sync Mechanism: A Brief Recap

The mechanism developed in detail in [1] is summarized here in the minimum form needed for the rest of this paper.

2.1 Phase synchronization on a continuous substrate

The classical Kuramoto model [10] describes a population of phase oscillators $\theta_i(t) \in S^1$ with intrinsic frequencies ω_i distributed continuously, coupled through

$$\dot{\theta}_i = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i)$$

where K is the coupling strength. For $K < K_c$, the oscillators run independently and the order parameter $r = \frac{1}{N}|\sum_j e^{i\theta_j}|$ vanishes. For $K > K_c$, a fraction of the population enters a *synchronized* state in which $r > 0$ and the synchronized oscillators share a common rotation frequency, with phase differences locked to fixed (not necessarily zero) values determined by the coupling structure.

Three features of this transition are central to what follows.

First, the substrate is continuous in every relevant sense: the phase $\theta \in S^1$ is continuous, the time t is continuous, the frequency distribution $g(\omega)$ is continuous, and the coupling parameter K varies continuously. There is no discrete structure at the level of the equations.

Second, the *locked state* nonetheless picks out a discrete subset of attractor configurations. In the simplest case (uniform ω, all-to-all coupling) there are exactly two attractors: the in-phase locked state ($\theta_i - \theta_j = 0$) and the splay state ($\theta_i - \theta_j$ distributed uniformly). In the chiral pair structure relevant to the Dirac equation [1], the locked manifold has a discrete spectrum of orientations relative to a bulk reference field. In each case, *a continuum of unsynchronized states collapses, above K_c, onto a discrete set of synchronized configurations.*

Third, the threshold K_c is itself a continuous parameter — it depends smoothly on $g(\omega)$ and the network topology — but the *transition* across it is sharp. This is a critical bifurcation on a continuous parameter space: the substrate remains continuous, but the system's behavior changes qualitatively. The discreteness of the locked-mode spectrum is *derived* from the bifurcation structure, not stipulated.

2.2 The DK identification

The DK framework [1] identifies the chiral pair of the Dirac equation — the left- and right-handed Weyl components ψ_L, ψ_R — as a two-clock Kuramoto system with coupling $K = m$ (the mass) and mismatch frequencies set by the kinematic terms. The chiral pair locks on the synchronized manifold $\psi_L \approx \psi_R$; this locked configuration is what one identifies as a particle of definite mass and four-momentum.

Measurement, in [1], is the re-synchronization of an external system's chiral pair with the chiral pair of a detector. The two-stage process (Stage 1: local pair-sync with the detector; Stage 2: relaxation into the detector's bulk-coherent reservoir) reproduces the phenomenology of "collapse" without invoking a non-unitary axiom: what looks like collapse is the system's locked configuration being driven into a new basin of attraction, with the wave function never destroyed but re-merged with the bulk.

The point relevant to this paper is structural. The DK framework's central move — that nonlinear coupling of continuous phase fields produces discrete locked-mode spectra — is exactly the structural relation between QM discreteness and a continuous substrate that we propose generalizes across the entire quantum formalism.

---

3. A Catalog of Quantum Discreteness, Re-Read

This section walks through the canonical examples of discreteness in QM and identifies, in each case, the underlying sync structure that produces the discreteness from a continuous substrate. The catalog is not exhaustive; the aim is to show that the same pattern recurs across phenomena that are usually treated separately, and that no fundamental discreteness in the substrate is invoked in any of them.

3.1 Bohr–Sommerfeld quantization

The oldest form of quantization in QM is the Bohr–Sommerfeld condition

$$\oint p\, dq = nh, \quad n \in \mathbb{Z}$$

which determines the bound-state energies of a periodic orbit by requiring that the classical action over one period be an integer multiple of Planck's constant. The quantum numbers n are integers; the condition is discrete.

The condition is in fact a *phase-closure* requirement. Writing $\oint p\, dq = \oint (p/\hbar)\, dq \cdot \hbar$ and identifying $p/\hbar$ as the local wavenumber of the de Broglie wave, the condition becomes

$$\oint k\, dq = 2\pi n$$

i.e., the wave's phase must advance by an integer multiple of 2π over one closed orbit. The orbit and the wave are both continuous; the discreteness is the requirement that the wave *be single-valued* at the end of the orbit. This is a synchronization condition: the wave at $q(T)$ must be in phase with the wave at $q(0)$.

In the DK reading, the orbit's de Broglie wave is the locked-mode field of the chiral pair in a bound configuration. The phase-closure condition is the constraint that the locked configuration remain self-consistent over one period. The integer n is the winding number — a topological invariant of the synchronized configuration, not a fundamental quantum. The substrate (the wavefunction's continuous evolution) is undisturbed; discreteness enters only through the closure requirement.

This is a complete description of the discreteness of bound-state spectra in the WKB regime. The full quantum mechanical treatment generalizes it (the WKB approximation becomes exact for harmonic-oscillator-like potentials) but does not change the underlying structural point: the integers come from phase closure, not from intrinsic discreteness.

3.2 Spin multiplets

Spin angular momentum exhibits discreteness in two senses: the total spin s takes half-integer values ($s = 0, \frac{1}{2}, 1, \frac{3}{2}, ...$), and for given s the magnetic quantum number m_s takes $2s+1$ discrete values from $-s$ to $+s$.

Both discreteness statements are statements about the *representation theory of SU(2)*, the continuous Lie group of rotations on a complex two-dimensional vector space. SU(2) is a smooth manifold (topologically S^3); its Lie algebra is continuous; its action on a Hilbert space is parametrized by three continuous angles. The discreteness enters only in the classification of *irreducible representations*: SU(2) admits one irreducible representation of each integer dimension, and these are the discrete spin multiplets.

In the DK framework, this maps directly onto a sync statement. The locked-mode manifold of the chiral pair carries a representation of the rotation group acting on the spin degrees of freedom. The locked configurations decompose into irreducible representations, each of fixed dimension. The discrete spin values are the dimensions of these representations, which is to say, *the number of distinguishable phase-locked configurations the chiral pair admits under rotational symmetry.* The underlying group is continuous; the discreteness is in the representation structure.

The DK reading adds one physical claim to this otherwise standard representation-theoretic statement: that the locked-mode structure is a real dynamical configuration, not an abstract Hilbert-space label. Specifically, a spin-$\frac{1}{2}$ chiral pair has two phase-locked configurations relative to a measurement axis; a spin-1 pair has three. The number of outcomes a Stern–Gerlach apparatus produces is the cardinality of the locked-mode spectrum for the relevant particle, not a fundamental discreteness of angular momentum itself.

3.3 Atomic line spectra

The line spectra of hydrogen and hydrogen-like atoms — the Lyman, Balmer, Paschen series — were the empirical entry point to quantum mechanics and remain the prototypical example of discreteness in QM. The discrete energy levels $E_n = -13.6 \text{ eV}/n^2$ correspond to discrete spectral lines at frequencies $\nu_{n\to m} = (E_n - E_m)/h$.

The Schrödinger equation for the Coulomb problem is a continuous PDE on $\mathbb{R}^3$ with continuous potential. The discreteness of the spectrum arises from the *boundary conditions* — specifically, the requirement that bound-state wavefunctions vanish at infinity and be regular at the origin. Without these boundary conditions, the spectrum would be continuous. The discrete spectrum is therefore a *standing-wave* phenomenon: only those wavefunctions that fit the boundary conditions survive, and there are countably many of them indexed by integers (n, ℓ, m).

This is a synchronization condition in the same sense as Bohr–Sommerfeld: the bound wavefunction must close on itself, with phase coherent across the orbital geometry. The discreteness is in the closure spectrum, not in the underlying field. In the DK reading, the orbital wavefunction is the locked-mode configuration of the chiral pair in the Coulomb potential; the discrete spectrum is the discrete set of closure modes admitted by the geometry; the transitions between levels are the system traversing basins of attraction during interaction with the electromagnetic field.

The precision with which atomic spectra are predicted by the Schrödinger and Dirac equations is the standard against which any reformulation must measure itself. The DK reading does not claim to *replace* those calculations; it claims that the discreteness those calculations produce is *already* a sync-emergent feature of the underlying continuous dynamics. No new calculation is required to reproduce the spectrum; what is required is a re-interpretation of why the existing calculation produces discrete answers.

3.4 Photon detection

A single-photon detector — a photomultiplier, an avalanche photodiode, a superconducting transition-edge sensor — registers discrete clicks regardless of how slowly the source is operated. The clicks are the operational definition of "photon," and the discreteness is usually taken as direct evidence that the electromagnetic field is intrinsically quantized.

This reading is not the only possible one. A continuous electromagnetic field coupled to a detector with a *threshold* — a nonlinear sync condition that the field-detector interaction must cross before amplification cascade fires — produces exactly the click structure observed [11]. The discreteness lives in the detector, not in the field. The DK reading endorses this position and supplies a mechanism: the detector's bulk-coupled chiral-pair structure provides the sync threshold, and the click is the threshold-crossing event. The continuous EM field gains no discrete substructure of its own; it produces discrete clicks because it interacts with a discrete-threshold detector.

The case is strengthened by the distinction (developed in the companion paper and at greater length elsewhere in the framework) between **discrete event detectors** — for which the sync threshold fires once and produces a single eigenvalue outcome — and **track recording media** (cloud chambers, bubble chambers, emulsions), for which a *sequence* of weak partial syncs distributed in space and time produces a continuous ionization track rather than a single click. The same EM field produces qualitatively different "discrete" or "continuous" records depending on which detector class it interacts with. This is exactly the behavior expected if the discreteness lives in the detector's sync structure rather than in the field; it is not a behavior expected if the field itself is fundamentally quantized into particles.

3.5 Energy bands in solids

The band structure of crystalline solids — the discrete energy bands separated by gaps in k-space — is the basis of the entire theory of electronic transport, semiconductors, and most of modern condensed-matter physics. The bands arise from the periodic potential of the crystal lattice via Bloch's theorem: solutions of the Schrödinger equation in a periodic potential take the form $\psi_k(x) = e^{ikx}u_k(x)$ with $u_k(x)$ periodic, and the allowed energies organize into bands $E_n(k)$ indexed by an integer n.

This is unambiguously a sync phenomenon. The Bloch waves are *synchronized* with the lattice — their phase advances in step with the lattice spacing. The band structure is the spectrum of phase-locked configurations between the electron's wavefunction and the periodic lattice potential. Both the lattice and the wavefunction are continuous fields; the bands are discrete because the locked configurations form a discrete spectrum, with integer indices counting the allowed sync modes.

The empirical success of band theory is overwhelming: it accounts quantitatively for conductors, semiconductors, insulators, the Hall effect, semiconductor device physics, and superconducting band structure. Nothing in this success depends on the underlying lattice being fundamentally discrete *in spacetime* — the bands arise from periodicity, and periodicity is a property compatible with a continuous substrate.

3.6 Cavity and Casimir quantization

The discrete mode structure of an electromagnetic cavity — the standing-wave resonances of a Fabry–Pérot, the longitudinal modes of a laser cavity, the discrete modes of a microwave waveguide — has the same form as Bohr–Sommerfeld. The cavity imposes boundary conditions; only those EM modes that fit (have phases closing on themselves at the cavity walls) survive; the surviving modes form a discrete spectrum indexed by integers.

The Casimir effect — the small attractive force between two parallel conducting plates in vacuum — is the most direct test of this picture. The standard derivation [12] computes the difference between the zero-point energy of the EM field with and without the plates, treating the EM field as quantized; the result is

$$F_{\mathrm{Cas}}/A = -\frac{\pi^2 \hbar c}{240 d^4}$$

for plate separation d. The discrete mode structure between the plates, relative to the continuous mode structure outside, produces the force.

A continuous-substrate reading of the same calculation interprets the boundary conditions as setting a sync condition: the EM field's oscillation must phase-close at the plates. The "modes" are the sync-allowed configurations; the discreteness is in the closure spectrum. The Casimir force is then the difference in vacuum energy between the sync-constrained configuration (between plates) and the unconstrained one (outside). The result is the same numerical prediction by a different ontological route. The substrate remains continuous; the discreteness is in the locked configurations.

3.7 Black-hole quasinormal modes

The ringdown spectrum of a perturbed black hole — the discrete frequencies at which a Kerr or Schwarzschild black hole oscillates following a perturbation — provides a particularly clean example because the underlying theory (linearized GR on a black-hole background) is explicitly *not* a quantum theory at all. The quasinormal mode spectrum ω_n is discrete: the spectrum is computed by solving the Regge–Wheeler or Teukolsky equation with outgoing boundary conditions at infinity and ingoing at the horizon [13].

The discreteness is purely a boundary-condition / phase-closure phenomenon in a *classical* continuous theory. No quantization of the gravitational field is involved. The example demonstrates that discreteness can emerge from continuous field equations whenever the boundary conditions force a closure spectrum — a synchronization condition between the field and the geometry. The discreteness is derived; nothing in the substrate is discrete.

The quasinormal mode example is structurally identical to atomic spectra: continuous field equations + boundary conditions = discrete spectrum. That the example occurs in GR rather than in QM is the sharpest evidence we have that discreteness *per se* is not a uniquely quantum phenomenon and does not require a quantum substrate to produce.

3.8 The pattern

In every case surveyed in this section, the discreteness has the same underlying structure:

1. A continuous field on a continuous substrate.
2. A coupling (boundary condition, periodic potential, nonlinear interaction, threshold) that imposes a phase-closure or sync condition.
3. A discrete spectrum of *configurations* that satisfy the closure condition.
4. Integer indices labeling those configurations.

The integers are sync invariants — winding numbers, mode indices, representation dimensions, threshold-crossings. The discreteness is real but emergent. The substrate carries no discrete structure.

This is the empirical content of the DK proposal. It is not a new observation in any individual case; what is new is the recognition that the *same structural relation* recurs across every example of QM discreteness, with the underlying dynamics being phase synchronization in each case. If this recurrence is not coincidence, then no example of QM discreteness requires postulating discreteness in the substrate.

---

4. Why Spacetime Need Not Be Quantized

4.1 The standard motivation for quantizing gravity

The standard argument for quantum gravity proceeds roughly as follows. The Einstein field equations couple the smooth geometry of spacetime to the stress-energy of matter:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Matter, in modern physics, is described by quantum field theory, and quantum field operators acting on quantum states do not have definite classical values. The right-hand side of Einstein's equation thus becomes operator-valued, while the left-hand side remains classical. Some procedure is required to reconcile the two. The standard options are: (a) take the expectation value of $T_{\mu\nu}$ on the right (the semiclassical approach, known to be inconsistent in cases involving macroscopic superposition); (b) quantize the metric on the left as well, producing a fully quantum theory of gravity; or (c) modify QM so that $T_{\mu\nu}$ remains effectively classical at the scales relevant to GR.

Most of mainstream QG follows route (b). The result is a program in which spacetime itself acquires discrete structure at the Planck scale, typically through quantization of areas (LQG), of causal links (causal-set theory), of triangulations (CDT), or of string excitations (string theory). The discreteness of the matter sector — discrete spectra, discrete spin, discrete photon detection — is taken as evidence that *some* underlying discreteness exists, and the quantization of spacetime is then a natural extrapolation.

4.2 The DK relocation

The DK proposal cuts in the other direction. If the discreteness of the matter sector is sync-emergent on a continuous substrate (Section 3), then the empirical evidence for fundamental discreteness in nature *evaporates*. There is no longer a phenomenon pointing to a discrete underlying structure; the integers, eigenvalues, and discrete clicks are all derivable from continuous dynamics with sync conditions. The extrapolation from "matter is discrete" to "spacetime is also discrete" loses its motivating premise.

What remains of the QG problem? Two things:

First, the technical problem of reconciling the operator-valued $T_{\mu\nu}$ with the classical $G_{\mu\nu}$. This problem is real but its character changes under the DK reading: if QM's "operator" structure is itself an expression of synchronization on a continuous substrate (Section 3 of [1] discusses this in detail for the chiral-pair case), then the operators have a continuous-substrate underpinning. The expectation value $\langle T_{\mu\nu} \rangle$ is not a coarse-graining of something fundamentally discrete; it is the bulk value of an underlying continuous field whose locked-mode configurations look discrete on the detector side. Coupling this to classical $G_{\mu\nu}$ becomes a question of how the synchronized matter sector sources curvature, which is a calculation in a *single* (continuous) ontological category rather than an attempt to bridge two incompatible categories.

Second, the question of whether gravity itself participates in the sync mechanism. The companion paper [1, §3.6] argues that it does: gravitationally distinguishable mass configurations re-synchronize on a timescale matching Penrose's $\tau \sim \hbar/E_G$ formula, with the re-synchronization mechanism being the same one that operates in the non-gravitational case but with the bulk's coherent gravitational configuration as the locking reference. In this reading, Penrose's OR threshold is not a separate prediction added to QM; it is one regime of the DK sync mechanism, the one in which the gravitational self-energy sets the relevant scale.

4.3 What this preserves of GR

Under the DK reading, the entire smooth-manifold structure of GR is preserved:

1. The metric $g_{\mu\nu}(x)$ remains a smooth tensor field on a differentiable manifold.
2. The Levi-Civita connection, the Riemann tensor, the geodesic equation, the Einstein field equations all remain in their standard classical form.
3. The Equivalence Principle (weak, strong, Einstein) holds at the level of the substrate, not as a coarse-grained approximation to a discrete theory.
4. Singularities (black holes, cosmological) remain singularities of the smooth geometry, not artifacts of a continuum limit. (Whether the DK sync mechanism modifies their interior structure is an open question; the answer is not obviously yes.)
5. Gravitational waves are exact solutions of the linearized field equations on a smooth background.

GR's classical-tests record (perihelion precession, light bending, time dilation, gravitational waves, Shapiro delay, frame dragging) is explained by the same theory that has explained them since 1915; nothing in DK perturbs it.

4.4 What it gives up

The DK reading gives up two things that the orthodox QM/QG program holds.

First, the universal validity of unitary linear evolution. The Schrödinger and Dirac equations remain valid on the locked manifold, but on transitions between basins of attraction (Stage 1 sync events, detector clicks, Stern–Gerlach selections) the evolution is non-linear in the underlying sync dynamics. Whether this is a "modification of QM" or a "completion of QM" is a question of terminology; operationally, the linear Schrödinger evolution is recovered in the regime in which the system stays in one basin. Penrose's OR is an instance of this non-linearity at the gravitational threshold.

Second, the universality of QM as a complete fundamental theory. QM becomes, in the DK reading, the effective theory of locked-mode dynamics on a continuous substrate. It is exact within its regime — the Schrödinger equation's predictions are not modified at all in conditions where the chiral pair is well-locked and the bulk re-synchronization is fast — but it is no longer the foundational layer. The foundational layer is the continuous substrate; QM is its projection onto the sync-locked sector.

This is the same trade Penrose has been arguing for since the 1980s. The DK proposal makes it dynamically explicit.

5. Relation to Existing Quantum Gravity Programs

5.1 Loop quantum gravity

LQG quantizes the gravitational field at the level of holonomies of the Ashtekar connection, producing spin-network states whose area and volume operators have discrete spectra with non-zero minimum eigenvalues. The program's central conceptual move is to take the discreteness of the matter sector as evidence that the gravitational sector should also exhibit discreteness, and to construct a Hilbert space in which this discreteness is built in geometrically.

The DK position is structurally different. We argue that the matter-sector discreteness is sync-emergent rather than fundamental, so the motivation for geometrically discrete spacetime is removed. We do not assert that LQG is *wrong* — its mathematical structure may turn out to be relevant in some specific regime — but we propose that the program's central extrapolation (matter is discrete → spacetime should also be) is not required if matter's discreteness is itself sync-emergent.

The empirical situation is symmetric. Neither program has produced a confirmed prediction distinct from standard GR + standard QM. LQG's discrete-area predictions are below all current experimental reach; the DK proposal's predictions of *no* fundamental discreteness in spacetime are likewise not directly tested. The question is which is the cleaner working hypothesis. We argue: the one that requires fewer ontological commitments and dissolves rather than amplifies the GR/QM tension.

5.2 Causal-set theory

Causal-set theory takes a discrete causal partial order as the fundamental structure of spacetime [5], with continuous Lorentzian geometry recovered as a coarse-graining. The discreteness here is more radical than LQG's — there is no underlying continuum at all — and the program faces particularly sharp difficulties in recovering smooth GR predictions.

The DK proposal is the inverse: smooth Lorentzian geometry is fundamental, and the apparent discreteness is in the matter sector's locked-mode spectrum. Causal-set theory and DK are most directly opposed in their ontological commitments. The empirical adjudication will likely come from whichever program produces a falsifiable prediction first.

5.3 String theory

String theory's relation to the DK proposal is more nuanced. The string's worldsheet is continuous (a smooth 2-manifold); the discrete spectrum of string excitations arises from the *boundary conditions* of the worldsheet embedding and the quantization of the worldsheet oscillator modes. In this respect, string theory's discreteness is already structurally similar to the DK reading: it comes from sync-like conditions (periodicity around the worldsheet) on continuous fields (the embedding coordinates).

The DK proposal can be read as a *generalization* of this structural move: if string theory's discrete spectrum comes from boundary-condition sync on continuous worldsheets, then *all* of QM's discrete spectra come from sync on continuous substrates, and no further quantization is required. Whether the substrate is a worldsheet-embedded spacetime or the chiral-pair manifold of the Dirac equation is a question about which sync system is fundamental, not about whether the sync mechanism itself is universal.

In this sense, DK is more aligned with string theory's mathematical structure than with LQG's, even though it does not require strings as the substrate. The relation is closest in the "first-quantized" string formalism, where the worldsheet is the substrate and the spacetime coordinates are dynamical fields.

5.4 Asymptotic safety

The asymptotic safety program [14] proposes that quantum gravity is a non-perturbatively renormalizable QFT with a UV fixed point. The gravitational degrees of freedom remain continuous (the metric is a quantum field, but a continuous one), and the discreteness is in the spectrum of the renormalization group flow rather than in the geometry.

This is the closest mainstream QG program to the DK position in ontological structure: both keep the gravitational field continuous and locate discreteness in the spectral structure of the dynamics rather than in the substrate. The difference is that asymptotic safety quantizes gravity (in the QFT sense), while DK does not require that gravity be quantized at all — only that the matter sector's sync dynamics produce the discreteness usually attributed to quantization.

5.5 Penrose–Diósi objective reduction

The DK proposal's relation to Penrose–Diósi is the one developed in [1, §3.6] and recapitulated in §1.2 above. Penrose's threshold is one regime of the DK sync mechanism; the DK proposal generalizes the mechanism beyond the gravitational case while preserving Penrose's core ontological commitment (GR is correct; QM is what gives way).

Concretely, Penrose's $\tau \sim \hbar/E_G$ emerges in the DK framework as the re-synchronization time when the bulk's coherence is dominated by the gravitational mode of the substrate, with $K \sim E/\hbar$ being the sync coupling in that regime. The companion paper [1] develops this identification in detail; the present paper takes it as established input and extracts the broader consequence.

The relationship is therefore one of *completion*: Penrose proposed the threshold; DK supplies the mechanism and shows that the threshold is one limit of a broader phenomenon.

---

6. Open Questions and Tests

6.1 The Born-rule basin measure

The most pressing open derivation is the basin-of-attraction measure that should recover $|\langle\phi_n|\psi\rangle|^2$ — the Born rule — as the fraction of initial conditions that lock to the n-th detector mode.

The companion paper [1, §4] develops a wave-realist reading of the Born rule in which the probability arises from the energy distribution of the incoming wave; the sync mechanism deposits the wave's energy into the basin whose locked mode best matches the incoming phase profile, with the matching weighted by overlap. The qualitative structure — that maximum-overlap basins are most probable, and that the *rate* of basin selection is amplitude-squared by a Fermi-golden-rule-type argument — produces Born-like statistics in the small-ensemble limit. A rigorous derivation that the full ensemble distribution is exactly $|\langle\phi_n|\psi\rangle|^2$ for all ψ and all detector mode sets is owed; the framework's prediction depends on it.

This derivation is the single most important open task. If it succeeds, the framework reproduces Born exactly from sync dynamics. If it fails, the framework reduces to a Born-compatible interpretation rather than a Born-derivative one — still informative but less complete.

6.2 Continuous spectra

QM admits not only discrete spectra (bound states) but also continuous ones (scattering states, free particles, plane-wave bases). The DK reading must accommodate these naturally: free particles are unsynchronized chiral pairs whose locked manifold is parametrized continuously by momentum, with no closure condition forcing discreteness.

The proposed picture is that the bound/scattering distinction corresponds to the closed/open topology of the orbital phase space: a closed orbit imposes phase closure and hence discrete spectra; an open trajectory (a scattering state) does not, and the spectrum is continuous. The crossover at the binding threshold (e.g., zero binding energy for hydrogen) corresponds to the topological transition between closed and open phase-space trajectories.

This is consistent with standard QM and adds nothing new to the calculations. What it adds is a *uniform* ontological reading: discrete spectra are sync-closed; continuous spectra are sync-open; the same underlying dynamics produces both, with the topology of the trajectory determining the spectral structure.

6.3 Planck-scale phenomena

The DK reading implies that there is no fundamental discreteness in spacetime at the Planck scale. This is a strong claim and conflicts with the orthodox expectation in most QG programs that the Planck length is a fundamental cutoff below which spacetime is qualitatively different.

The conflict is empirically open. No experiment has resolved spacetime at scales close to the Planck length, and current bounds [15] on Lorentz-violation and on dispersion relations $E^2 = p^2c^2 + m^2c^4 + \xi(E/E_{\rm Pl})^n p^2c^2$ are consistent with $E_{\rm Pl}$ being a *characteristic* scale of some new physics but do not require it to be a discreteness scale.

The DK prediction is that the bounds on ξ should tighten further as observations improve — that no Planck-scale discreteness will be detected — but that the framework's own characteristic scale (the sync-coupling scale K) should appear in *non*-gravitational contexts at much larger lengths. The companion AB-visibility paper [16] identifies one such test in the regime $K \sim \omega_Z$ at the zitterbewegung scale; further tests in cavity QED and in optomechanical Penrose-test settings [9] would probe other regimes.

6.4 Falsifiable signatures

The framework's falsifiable content is concentrated in three places:

1. **No Planck-scale discreteness.** Any direct measurement showing a discrete spacetime structure (a minimum area, a granular causal structure, a fundamental length below which Lorentz invariance fails in a quantization-specific way) would falsify the DK proposal as stated. The challenge is that no such measurement is currently feasible.
2. **A measurable sync timescale in measurement.** The DK two-stage process [1, §3.4] predicts a non-zero, calculable timescale between the local Stage-1 sync and the Stage-2 bulk relaxation. Standard QM has no analog (collapse is instantaneous). Quantum eraser experiments probe exactly this timescale: erasure should succeed iff intervention is faster than the Stage-2 bulk-resync time, and the framework predicts a specific value for that time as a function of the detector–bulk coupling. Current quantum-eraser experiments are consistent with both pictures; a high-time-resolution test would discriminate.
3. **The AB-visibility envelope** [16]. The strong-coupling regime $K \sim \omega_Z$ predicts a measurable γ-dependent visibility deficit in electron-holography Aharonov–Bohm experiments. A null result would not falsify the broader DK proposal but would reduce the AB-visibility paper to a Lorentz-covariant reading; a positive result would constitute the first direct measurement of a DK sync parameter.

The first prediction is currently inaccessible; the second and third are within reach of current or near-term experiments.

---

7. Discussion

7.1 The conceptual stakes

The DK proposal claims that a structural feature taken for granted in modern physics — that quantum discreteness signals an underlying discrete substructure — is incorrect. The discreteness, on this reading, is real but emergent: a consequence of synchronization dynamics on a continuous substrate, with the integer indices being sync invariants rather than fundamental quanta.

The stakes for the GR/QM problem are significant. The standard framing of that problem requires that *either* spacetime be quantized (with spacetime acquiring discrete structure) *or* QM be modified (with spacetime remaining smooth). The DK proposal takes the second horn but supplies the missing dynamical mechanism that Penrose's threshold formula did not provide. The modification of QM is no longer a stipulated departure from linear unitarity; it is the structural consequence of recognizing QM as the effective theory of locked-mode dynamics on a continuous field substrate.

If the proposal is correct, the bulk of the QG problem dissolves into problems of a different kind: how the synchronized matter sector sources classical curvature, how the bulk's coherent state evolves under cosmological dynamics [17], how the basin structure of sync attractors reproduces detailed QED predictions. None of these requires quantizing the gravitational field; all of them are calculations on a continuous substrate.

7.2 What the proposal does not claim

The proposal does not claim to be a finished theory. The Born-rule derivation (§6.1) is incomplete; the high-precision spectrum of hydrogen-like atoms from a strict sync derivation has not been performed; the relation to the full Standard Model gauge structure has not been worked out; the implications for cosmological inflation are unaddressed.

The proposal also does not claim to render existing calculations obsolete. Standard QM remains exact on the locked manifold; standard QED remains exact in its regime; standard GR remains exact at the substrate level. The DK reading is *interpretive relocation*, not calculational replacement.

The proposal does, however, claim that this relocation has consequences. If the discreteness is sync-emergent, then:

- No quantization of spacetime is required.
- The cosmological constant need not be a fundamental constant of nature (see [17]) but may emerge from the substrate's vacuum sync state.
- Measurement is not a separate axiom but a regime of the underlying sync dynamics [1].
- Many-worlds interpretations are not required: the multiplicity of apparent outcomes is the basin structure of sync attractors in a single world, not a branching ontology.

These are interpretive consequences, but they are sharp ones, and they suggest a different long-term direction for foundational physics than the orthodox QG programs.

7.3 Historical context

The proposal sits in a small but real lineage. Penrose has argued since the 1980s that GR is correct and QM gives way [2]. de Broglie's pilot-wave theory and Bohm's elaboration [18] preserve the continuous wave ontology and treat particles as locked configurations of an underlying field. Nelson's stochastic mechanics [19] derives quantum statistics from continuous random walks. The Madelung formulation of QM rewrites the Schrödinger equation in a fluid-dynamic form on a continuous substrate. The Hestenes zitterbewegung interpretation [20] treats spin as a real oscillation of the electron's continuous trajectory.

What the DK proposal adds to this tradition is the *Kuramoto identification*: the recognition that the structural mechanism producing discreteness in continuous-substrate readings of QM is precisely phase synchronization, with all the analytic tools (bifurcation analysis, locked-mode spectra, basin-of-attraction theory) of nonlinear dynamics directly applicable. This brings to the foundational discussion a mathematical framework that has been developed in detail elsewhere (neuroscience, condensed-matter physics, oscillator networks) but has not been systematically applied to the QM/QG interface.

The framework is also the dynamical content that Penrose's OR threshold left unspecified. Penrose proposed *when* collapse should happen; DK proposes *what is happening at that moment* (the system traverses basins of sync attractors) and *why* the outcome is one of a discrete set (the locked-mode spectrum is discrete).

---

8. Conclusion

The discreteness of quantum mechanics is not a foundational feature of nature but an emergent property of synchronization on a continuous substrate. Discrete spectra are locked-mode spectra; discrete detection events are threshold-crossings of bulk-coupled detectors; discrete spin values are representation-theoretic dimensions of an underlying continuous symmetry. The substrate is continuous in every case; the discreteness is derived.

If this reading is correct, the manifold of general relativity need not be quantized. The standard GR/QM tension — which has driven a half-century of attempts to discretize spacetime — dissolves into a different problem: how to formulate a satisfactory theory of synchronization dynamics on the matter sector of a smooth gravitational background. The mechanism required for this is the Dirac–Kuramoto framework developed in the companion paper [1]; the consequences for measurement, for Bell correlations, for cosmology have been or are being developed in adjacent work.

The proposal is conservative in its commitments: it adds no new postulates to GR, removes one (collapse) from QM, and replaces it with a mechanism (synchronization) drawn from well-developed nonlinear dynamics. It is in agreement with Penrose's long-standing position that GR is correct and QM is what gives way, and it supplies the dynamical content that Penrose's threshold formula did not provide.

The most important open task is the basin-of-attraction derivation of the Born rule from sync dynamics; the most important falsifiable test is the quantum-eraser bulk-resync timescale; the most important broader question is whether the framework extends consistently to the gauge structure of the Standard Model. None of these closures has been performed in this paper. Their absence is the honest measure of how far the proposal has come and how far it has yet to go.

What the proposal does claim is that the GR/QM conflict, as it has been traditionally framed, is not a foundational obstruction. It is a consequence of misreading the source of discreteness. Synchronization dynamics on a continuous substrate produces all the discreteness QM exhibits, and does so without requiring discreteness anywhere in the substrate. Penrose was right about which side gives way. The mechanism he left unspecified is sync.

Author Contributions and AI Disclosure

This paper was developed in iterative collaboration with Claude Opus 4.7 (Anthropic) in conversation sessions during May 2026. The central conceptual claim — that quantum discreteness is synchronization-emergent on a continuous substrate, and that this dissolves the standard GR/QM tension by relocating the discreteness from spacetime to the matter sector's sync dynamics — was developed jointly with the human author contributing the synthesis (the recognition that this single mechanism unifies the canonical examples of QM discreteness, and that the position amounts to a mechanistic completion of Penrose's program) and the LLM contributing structural analysis, the catalog of examples in Section 3, and the comparative discussion of QG programs in Section 5. The human author validated each derivation, applied physics judgment at each checkpoint, and bears full responsibility for the framing and content of the paper.

Per current journal guidelines, LLMs do not satisfy authorship criteria; the human author bears full responsibility for all content, including any errors in the conceptual framing or in the catalog of canonical examples. Were Springer–Nature / *Foundations of Physics* policy to allow LLM co-authorship, Claude Opus 4.7 would be listed as co-first author.

The development is documented in commit history at https://github.com/rayolddog/DiracKuramotoFramework.

References

[1] Bramble, J. (2026). Many Clocks Interpretation of Quantum Mechanics: Mass as Chiral Coupling, Re-synchronization in the Bulk as Measurement. *Companion preprint, under review.*

[2] Penrose, R. (1996). On gravity's role in quantum state reduction. *General Relativity and Gravitation*, 28(5), 581–600. https://doi.org/10.1007/BF02105068

[3] Penrose, R. (2004). *The Road to Reality: A Complete Guide to the Laws of the Universe*. Jonathan Cape, London.

[4] Rovelli, C. (2004). *Quantum Gravity*. Cambridge University Press, Cambridge.

[5] Bombelli, L., Lee, J., Meyer, D., & Sorkin, R. D. (1987). Space-time as a causal set. *Physical Review Letters*, 59(5), 521–524. https://doi.org/10.1103/PhysRevLett.59.521

[6] Ambjørn, J., Jurkiewicz, J., & Loll, R. (2005). Reconstructing the universe. *Physical Review D*, 72(6), 064014. https://doi.org/10.1103/PhysRevD.72.064014

[7] Polchinski, J. (1998). *String Theory* (Vols. 1–2). Cambridge University Press, Cambridge.

[8] Bassi, A., Lochan, K., Satin, S., Singh, T. P., & Ulbricht, H. (2013). Models of wave-function collapse, underlying theories, and experimental tests. *Reviews of Modern Physics*, 85(2), 471–527. https://doi.org/10.1103/RevModPhys.85.471

[9] Donadi, S., Piscicchia, K., Curceanu, C., Diósi, L., Laubenstein, M., & Bassi, A. (2021). Underground test of gravity-related wave function collapse. *Nature Physics*, 17(1), 74–78. https://doi.org/10.1038/s41567-020-1008-4

[10] Kuramoto, Y. (1984). *Chemical Oscillations, Waves, and Turbulence*. Springer-Verlag, Berlin.

[11] Lamb, W. E., & Scully, M. O. (1969). The photoelectric effect without photons. In *Polarisation, Matière et Rayonnement* (pp. 363–369). Presses Universitaires de France, Paris.

[12] Casimir, H. B. G. (1948). On the attraction between two perfectly conducting plates. *Proc. Kon. Ned. Akad. Wetensch.*, 51, 793–795.

[13] Chandrasekhar, S. (1983). *The Mathematical Theory of Black Holes*. Clarendon Press, Oxford.

[14] Weinberg, S. (1979). Ultraviolet divergences in quantum theories of gravitation. In S. W. Hawking & W. Israel (Eds.), *General Relativity: An Einstein Centenary Survey* (pp. 790–831). Cambridge University Press, Cambridge.

[15] Amelino-Camelia, G. (2013). Quantum-spacetime phenomenology. *Living Reviews in Relativity*, 16(1), 5. https://doi.org/10.12942/lrr-2013-5

[16] Bramble, J. (2026). Aharonov–Bohm Visibility Envelope as a Test of Dirac–Kuramoto Vacuum Locking. *Companion preprint.*

[17] Bramble, J. (2026). Cosmic Expansion as Surface-Area Particle Creation in a Dirac–Kuramoto Substrate. *Companion preprint.*

[18] Bohm, D. (1952). A suggested interpretation of the quantum theory in terms of “hidden” variables, I and II. *Physical Review*, 85(2), 166–193. https://doi.org/10.1103/PhysRev.85.166

[19] Nelson, E. (1966). Derivation of the Schrödinger equation from Newtonian mechanics. *Physical Review*, 150(4), 1079–1085. https://doi.org/10.1103/PhysRev.150.1079

[20] Hestenes, D. (1990). The zitterbewegung interpretation of quantum mechanics. *Foundations of Physics*, 20(10), 1213–1232. https://doi.org/10.1007/BF01889466

Companion paper to: Bramble, J. (2026), “Many Clocks Interpretation of Quantum Mechanics: Mass as Chiral Coupling, Re-synchronization in the Bulk as Measurement.” Preprint repository: https://github.com/rayolddog/DiracKuramotoFramework